Name of Candidate: _____ Form 4 _____

KINGS FOUNDATION PVT SEC SCHOOL

(The ladder to success)

2017 MSCE MOCK EXAMINATIONS

MATHEMATICS

Subject number M131/1

Thursday March 30th, 2017

Time allowed: 2 hours

07:30 – 09:30AM

PAPER I
(100 MARKS)

Instructions

- This paper contains **12** pages. Please check.
- **Answer all the 20 questions in this paper in the spaces provided below each question**
- The maximum number of marks for each question is indicated against each question
- Electronic calculators may be used
- All working must be clearly shown
- Write your name and class at the top of each page of your question paper in the spaces provided
- **Marks will be lost for untidy work**

Question Number	Tick if answered	Do not write in these columns	
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			
13			
14			
15			
16			
17			
18			
19			
20			
TOTAL			

© KINGS MOCK EXAMINATIONS 2017

Turn over… /

Name of Candidate: _____ Form 4 _____

1. In a race, the probability that John or Paul wins is $\frac{1}{5}$. Given that the probability that John wins is $\frac{4}{25}$. Find the probability that Paul wins.

(3 marks)

2. Simplify $\frac{2(x^2-20)}{(x-2\sqrt{5})(x+2\sqrt{5})}$ leaving answer with a rational denominator.

(3 marks)

3. Solve the equation $\frac{4^x}{8} = (2^{x-1})\sqrt{2}$ (5 marks)

Name of Candidate: _____ Form 4 _____

4. Figure 1 shows the graph of Takondwa jumping out of a parachute

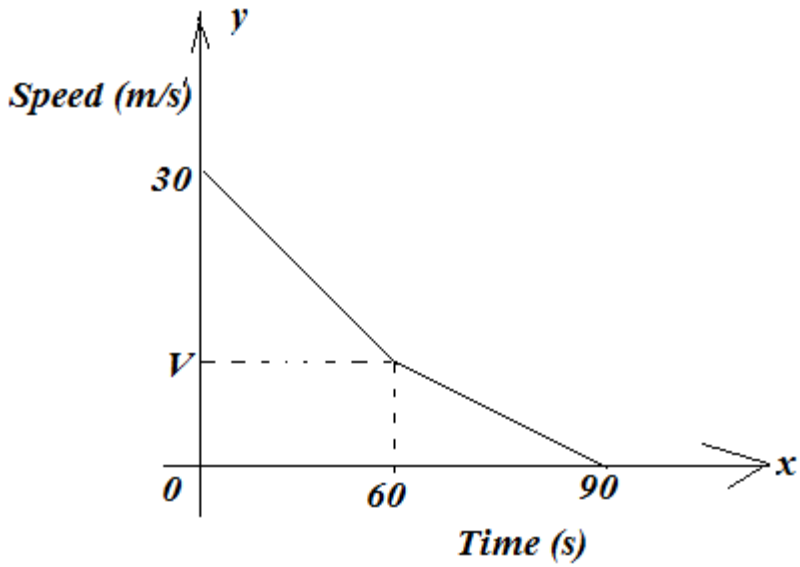


Figure 1

Given that the distance travelled in the first 60 seconds is 1140 metres, find the value of Vm/s

(4 marks)

5. Solve the logarithmic equation to find the value of x

$$\log_{\sqrt{x}} 16 - \log_{\sqrt{x}} 2 = 3$$

(5 marks)

Name of Candidate: _____ Form 4 _____

6. Given that $a + 3b$ is a factor of $2a^3 + 7a^2b - 9b^3$. Find the other two linear factors. (6 marks)

7. In figure 2 below ABC is a tangent to circle BEFD at point B. EDC and BF are straight lines. Angle BCE=35^0 , angle BED=40^0 and angle DEF =28^0

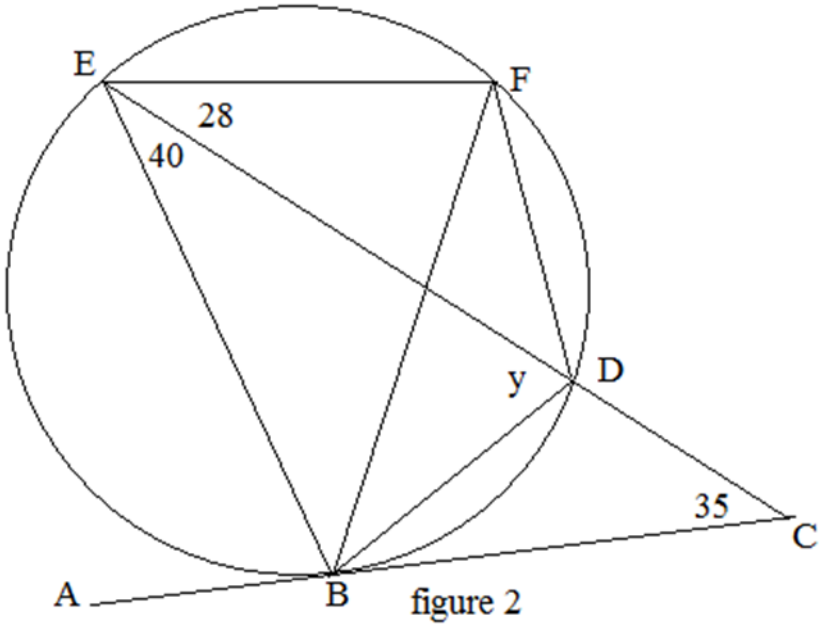


figure 2

Find the size of angle BFE (4 marks)

Name of Candidate: _____ Form 4 _____

8. Given that $h(x) = (x^2 - 8)^3$. Find the value of x if $h(x) = -8$

(4 marks)

9. Figure 3 below is an object made from a hemisphere and a cone of radius 3cm .The vertical height h and the slant height l of the cone makes an angle 60^0 at the vertex of the cone.

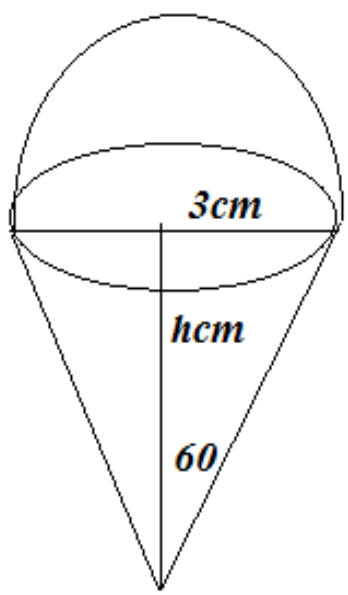


Figure 3

Calculate the total volume of the object taking π as $\frac{22}{7}$ (6 marks)

Name of Candidate: _____ Form 4 _____

10. Figure 4 below shows the graph of the equation $y = -x^2 - 2x + 8$.

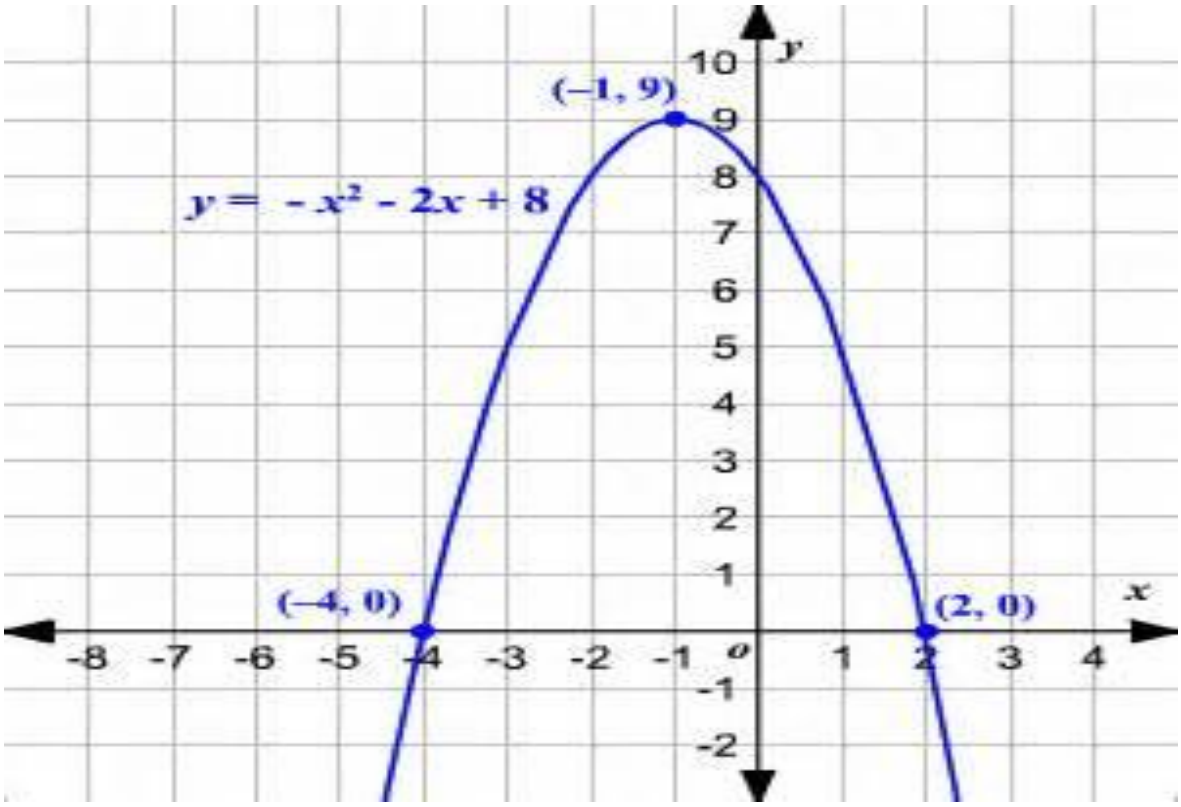


Figure 4

Use the graph to solve the simultaneous equations

$$y = -x^2 - 2x + 8 .$$
$$y = 2x + 8$$

(5 marks)

Name of Candidate: _____ Form 4 _____

11. Find the **standard deviation** of the discrete frequency distribution below

score (x)	**3**	**11**	**21**
frequency	**2**	**3**	**1**

(7 marks)

12. Figure 5 below shows a quadrilateral ABCD inscribed in a circle, ABP is an equilateral triangle with AB = AP = BP. Angle ADC is acute (less than 90 degrees

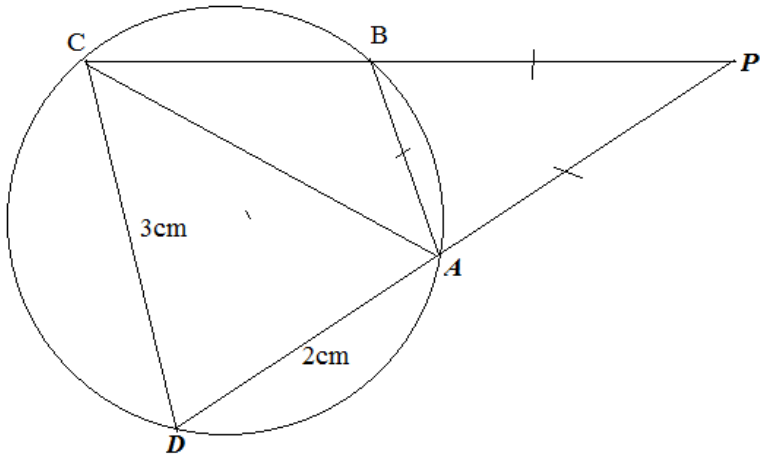


Figure 5

Calculate the area of triangle ADC leaving answer in surd form.

(5 marks)

Name of Candidate: _____ Form 4 _____

13. Make λ subject of formula $a\lambda^2 + b\lambda - h = 0$ (6 marks)

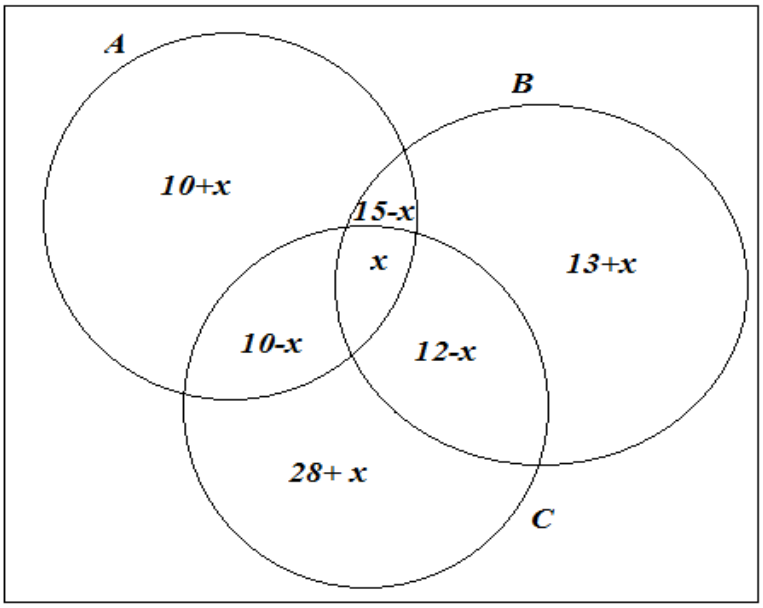
14. A boat sails 4km on a bearing of 038^0 and 5km on a bearing of 067^0.

Calculate the bearing of the boat from its starting point. (7 marks)

Name of Candidate: _____ Form 4 _____

15. The distance between two points $R(\chi, 4)$ and $S(7,9)$ is $5\sqrt{2}$ cm. Find the possible values of χ (6 marks)

16. The Venn diagram below show the number of elements in three sets A, B and C.



Given that $n(A \cup B \cup C) = 93$. Find $n(A \cap B)'$ (4 marks)

Name of Candidate: _____ Form 4 _____

17. Geoffrey is looking at a bird at the top of a straight pole. Geoffrey is 1.65 metres tall and the angle of elevation of the bird from Geoffrey is 60^0. How high is the pole if he is 30m away from the tree. (5 marks)

18. The second term of a geometric progression G.P is 12 and the 6th term is 60.75. Find the first term a (6marks)

Name of Candidate: _____ Form 4 _____

19. Table below shows the distribution of ages of workers at a certain factory in Blantyre.

AGE	**NUMBER OF WORKERS**	
20-29	18	
30-39	27	
40-49	36	
50-59	16	
60 or older	3	
TOTAL	**100**	

If a person is selected at random, find the probability that the person is **neither** 40 years **nor** older. (5 marks)

Name of Candidate: _____ Form 4 _____

20. Figure 6 below shows a parallelogram OABC in which $\overrightarrow{OA} = \underline{a}$ and $\overrightarrow{OC} = \underline{c}$

M is the midpoint of $\overrightarrow{AB}$

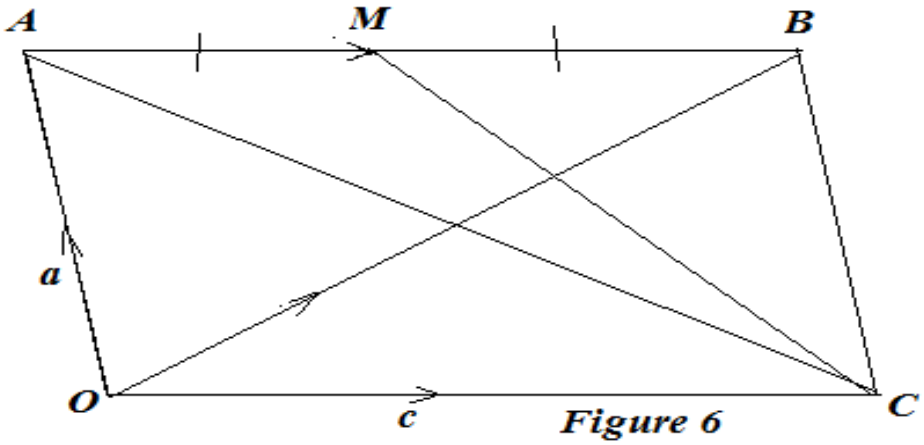


Figure 6

Express $\overrightarrow{MC}$ in terms of $\underline{a}$ and $\underline{c}$ **(4 marks)**

End of Question Paper